

Question

Under anhydrous conditions, PPh_3 and CCl_4 are reacted in a certain proportion to obtain an ionic compound **A**. The mass fraction of Cl in **A** is 11.7%. **A** is treated with $P(NMe_2)_3$ to obtain compound **B**. **B** does not contain nitrogen and the mass fraction of P is 11.6%. In the structure of **B**, all bond angles centered on carbon atoms are close to 120° . **B** reacts with an equimolar amount of $AuCl$ to obtain complex **C**, and **C** reacts further with an equimolar amount of $AuCl$ to obtain **D**. The mass fraction of Au in **D** is 39.3%. **B** is treated with 2 equivalents of n-butyllithium at low temperature and then reacted with $BeCl_2$ to obtain **E**, which contains 2 rings containing Be . In addition, **B** can also be used as a catalyst for certain organic reactions.

Regarding the above content, there are the following statements:

1. All carbon atoms in **A** have the same hybridization.
2. Theoretically, there are 4 sets of peaks in the $^{13}CNMR$ spectrum of **B**.
3. **B** can be stored normally in air.
4. **C** does not contain chlorine.
5. The coordination number of Au in **D** is 3.
6. When **B** is used as a catalyst, its function is more likely to be a Lewis acid than a Lewis base.
7. The coordination number of Be in **E** is 4.
8. The theoretical molar ratio of PPh_3 to CCl_4 when preparing **A** is 2 : 1.

Which of the following options corresponds to a set of completely incorrect statements?

A. All other options are incorrect

B. 1, 3, 5, 7

C. 1, 3, 6, 8

D. 3, 4, 5, 6

E. 1, 2, 7, 8

F. 2, 3, 5, 8

G. 2, 4, 6, 8

H. 2, 3, 4, 5

Answer

Correct Answer: **G**

Detailed Explanation

According to organic chemistry knowledge, PPh_3 and CCl_4 easily react to yield $Ph_3P = CCl_2$ and Ph_3PCl_2 . The latter is obviously not consistent, and the mass fraction of Cl in the former is too high. Considering its further addition with PPh_3 , the chemical formula of **A** can be obtained as $(Ph_3P)_2CCl_2$, and the mass fraction is also consistent with the question. Since it is an ionic compound, considering the electron-donating property of PPh_3 , its actual structure is $[(Ph_3P)_2CCl]^+Cl^-$.

CHECKPOINT

2 PTS

The structure of **A** is $[(Ph_3P)_2CCl]^+Cl^-$

$P(NMe_2)_3$ is a reducing agent and should reduce the central carbon atom of **A**. Therefore, **B** is $Ph_3P = C = PPh_3$, which also matches the mass fraction in the question.

CHECKPOINT

2 PTS

B is $Ph_3P = C = PPh_3$

C is $(Ph_3P)_2CAuCl$. According to the mass fraction, **D** is $(Ph_3P)_2C(AuCl)_2$.

CHECKPOINT

1 PTS

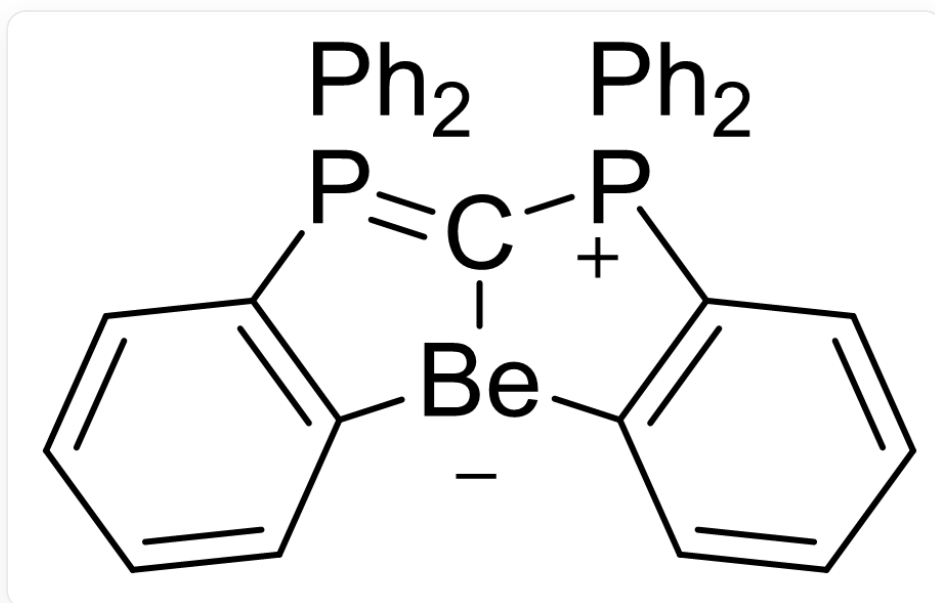
C is $(Ph_3P)_2CAuCl$

CHECKPOINT

1 PTS

D is $(Ph_3P)_2C(AuCl)_2$

n-Butyllithium abstracts one hydrogen each from the PPh_3 on both sides of **B**. Also note that the central carbon atom of **B** has a strong coordination ability. Therefore, **E** is

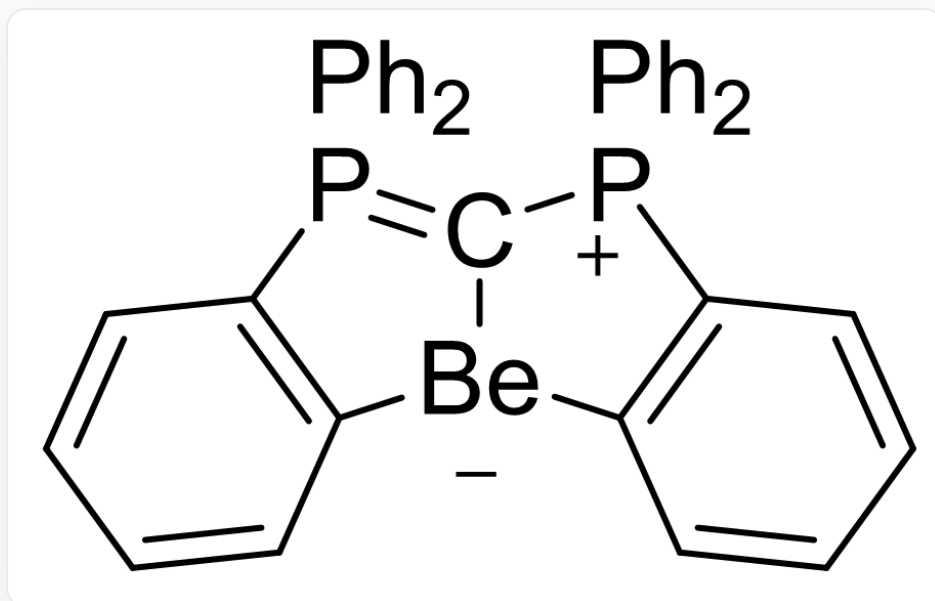


C1([Be-]2C(C=CC=C3)=C3[P+](C4=CC=CC=C4)1C5=CC=CC=C5)=P(C6=C2C=CC=C6)
(C7=CC=CC=C7)C8=CC=CC=C8

CHECKPOINT

3 PTS

E is



C1([Be-]2C(C=CC=C3)=C3[P+](C4=CC=CC=C4)1C5=CC=CC=C5)=P(C6=C2C=CC=C6)(C7=CC=CC=C7)C8=CC=CC=C8

All carbons in **A** are sp^2 hybridized. Statement 1 is correct.

There are 5 types of carbon in **B**. **B** is a strong Lewis base and is obviously sensitive to water and oxygen. Statements 2, 3, and 6 are incorrect.

C contains chlorine element. Statement 4 is incorrect.

In **D**, due to the presence of $Au - Au$ interaction, Au is 3-coordinate. Statement 5 is correct.

In **E**, Be is 3-coordinate. Statement 7 is incorrect.

The theoretical molar ratio of PPh_3 and CCl_4 in the preparation of **A** is 3 : 1. Statement 8 is incorrect.

CHECKPOINT

2 PTS

Statements 1 and 5 are correct, the rest are incorrect

Only the statement in option G is completely incorrect. Choose G.